Water quality testing

Klevis Imeri

The input to the problem:

$$\mathcal{I} := \left\{ \begin{array}{rll} N := \{\text{laboratories}\}, & |N| = n, \\ T := \{\text{contaminants}\}, & |T| = t, \\ M[i][j] := \{k \in T \mid \text{device } j \text{ from lab } i \text{ tests } k\}, & i \in N,\ j \in [m], \\ c[i][j] := \text{purchase cost of device } j \text{ from lab } i, & c[i][j] \geq 0, \\ d[i] := \text{delivery fee of laboratory } i, & d[i] \geq 0. \end{array} \right\}.$$

The output is a set of selected testing devices

$$\mathcal{S} := \{(i,j) \mid \text{we select device } j \text{ from laboratory } i\}.$$

$$\text{feasible}(\mathcal{S}) := \mathcal{S} \subseteq N \times [m] \ \wedge \bigcup_{(i,j)\in\mathcal{S}} M[i][j] = T.$$

The cost function is

$$f(\mathcal{S}) := \sum_{(i,j)\in\mathcal{S}} c[i][j] + \sum_{\substack{i\in N \\ \exists j\in[m]:\,(i,j)\in\mathcal{S}}} d[i].$$

Goal 1. For the natural LP given in the exercise, show that:

$$\forall B > 0,\ \exists \mathcal{I}: \quad |N| = |T| = 1 \ \wedge\ \mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}) > B\,\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}).$$

$$\begin{aligned} \mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}):&\ \text{minimum cost of a feasible selection of devices,} \\ \mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}):&\ \text{optimal value of the LP given in the exercise.} \end{aligned}$$

Fix an arbitrary $B > 0$. We construct an instance $\mathcal{I}_B$ depending on B, with one laboratory and one contaminant, such that

$$\mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}_B) > B\,\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}_B).$$

Choose one laboratory, one contaminant, and $m =?$ devices, each testing that contaminant. Set

$$d_1 = 1, \qquad c_{1j} = 0 \quad \forall j \in [m].$$

Since there is only one laboratory, the LP becomes

$$\begin{aligned} \text{minimize} \quad & d_1 y_1 + \sum_{j=1}^{m} c_{1j} x_{1j} = y_1 \\ \text{subject to} \quad & \sum_{j=1}^{m} x_{1j} \geq 1, \\ & x_{1j} \leq y_1 && \forall j \in [m], \\ & 0 \leq y_1, x_{1j} \leq 1 && \forall j \in [m]. \end{aligned}$$

To minimize y_1, we choose

$$x_{1j} = y_1 = \frac{1}{m} \qquad \forall j \in [m].$$

This is feasible. To see that a smaller cost is impossible, suppose $y_1 < 1/m$. Since $x_{1j} \leq y_1$ for every device,

$$\sum_{j=1}^{m} x_{1j} \leq m y_1 < 1,$$

contradicting the coverage constraint. Thus,

$$\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}_B) = \frac{1}{m}.$$

For the ILP, the variables are in $\{0, 1\}$. The coverage constraint requires at least one device with $x_{1j} = 1$, which forces $y_1 = 1$. Selecting just one device is feasible and costs 1, since all device costs are zero. Therefore,

$$\mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}_B) = 1.$$

Substituting these values into our target inequality gives

$$\mathrm{OPT}_{\mathrm{ILP}}(\mathcal{I}_B) > B\,\mathrm{OPT}_{\mathrm{LP}}(\mathcal{I}_B) \iff 1 > \frac{B}{m} \iff m > B.$$

Thus, choosing any integer $m > B$ proves our claim. □